



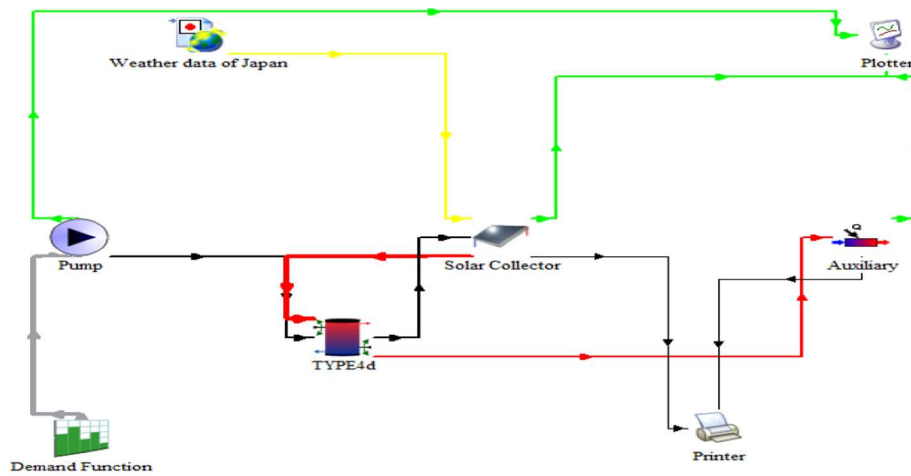
Project 2  
Solar Thermal Energy  
REE-401

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## The design:



This is the connection between the system elements which shows how does the system work after adding the storage on the same system that we have worked on in project1.

here are the values of our inputs and its connections.

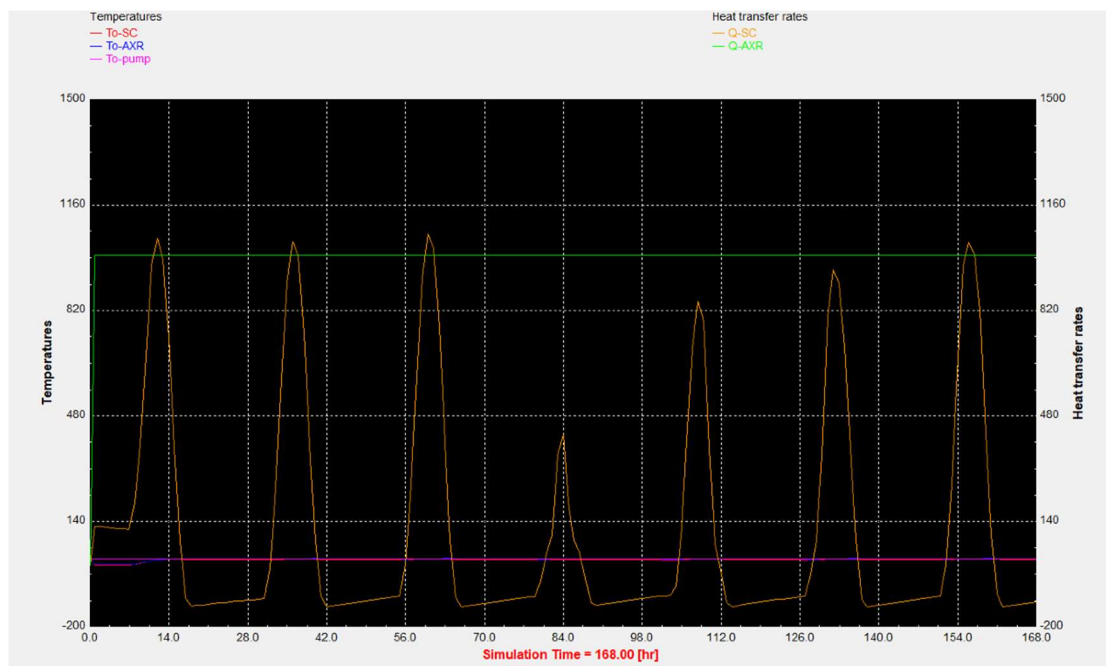
|                                                                                     |                                                                                     |      |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|
|  |  |      |
| Outlet fluid temperature                                                            | Hot-side temperature                                                                | 45.0 |
| Outlet flow rate                                                                    | Hot-side flowrate                                                                   | 0    |
| Power consumption                                                                   | Cold-side temperature                                                               | 15   |
|                                                                                     | Cold-side flowrate                                                                  | 0    |
|                                                                                     | Environment temperature                                                             | 20   |
|                                                                                     | Control signal for element-1                                                        | 1    |
|                                                                                     | Control signal for element-2                                                        | 0.0  |

|                                                                                     |                                                                                       |       |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------|
| project 1@) TYPE4d -> Solar Collector                                               |                                                                                       |       |
| Classic   Table                                                                     |                                                                                       |       |
| Select variable filter: All                                                         |                                                                                       |       |
|  |  |       |
| Temperature to heat source                                                          | Inlet temperature                                                                     | 20.0  |
| Flowrate to heat source                                                             | Inlet flowrate                                                                        | 100.0 |
| Temperature to load                                                                 | Ambient temperature                                                                   | 10.0  |
| Flowrate to load                                                                    | Incident radiation                                                                    | 0.    |
| Thermal losses                                                                      | Total horizontal radiation                                                            | 0.0   |
| Energy rate to load                                                                 | Horizontal diffuse radiation                                                          | 0.0   |
| Internal energy change                                                              | Ground reflectance                                                                    | 0.2   |
| Auxiliary heating rate                                                              | Incidence angle                                                                       | 45.0  |
| Element 1 power                                                                     | Collector slope                                                                       | 0.    |

|                                                                                     |                                                                                     |      |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|
|  |  |      |
| Outlet temperature                                                                  | Hot-side temperature                                                                | 45.0 |
| Outlet flowrate                                                                     | Hot-side flowrate                                                                   | 0    |
| Useful energy gain                                                                  | Cold-side temperature                                                               | 15   |
|                                                                                     | Cold-side flowrate                                                                  | 0    |
|                                                                                     | Environment temperature                                                             | 20   |
|                                                                                     | Control signal for element-1                                                        | 1    |
|                                                                                     | Control signal for element-2                                                        | 0.0  |

|                                                                                     |                                                                                       |       |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------|
|  |  |       |
| Temperature to heat source                                                          | Inlet fluid temperature                                                               | 20.0  |
| Flowrate to heat source                                                             | Fluid mass flow rate                                                                  | 100.0 |
| Temperature to load                                                                 | Control Function                                                                      | 1     |
| Flowrate to load                                                                    | Set point temperature                                                                 | 60.0  |
| Thermal losses                                                                      | Temperature of surroundings                                                           | 20.0  |

Here is the graph which shows the result of system.



## Economics

Now we are at the most important part for the companies and anyone who wants to invest in renewable energy systems especially solar thermal systems which are our case. The biggest question is : does it worth ?

In this section we are going to study the system economically which is crucial specially when it comes to renewable energy. In our case we are going to study solar thermal system and we are going to justify if the system worth it or not

### The storage tank

Storage tank of 1000 liter capacity costs about 500 usd.

### The pump

Pump with 10 kg/hr flow-rate is costs about 100 usd.

### The solar collector

We are working to calculate a  $1m^2$ . Area and efficiency of 0.8 and that costs around 500 usd.

## The auxiliary heater

We have selected one that has an average price of 200usd.

## The labor

The labor is an important part to take care of its price and it will cost around 100 usd.

## Insulation and piping

This aspect is essential for building a robust system and will cost approximately \$400.

## Calculation:

$$\text{Initial cost} = 500 + 100 + 500 + 200 + 100 + 400 + 1800\$$$

## . The maintenance

Maintenance plays a crucial role in preserving the system's efficiency and costs approximately 0.5% of the initial installation cost each year. This amounts to about \$18 annually, a relatively small expense considering its significance and value. The system has a lifecycle of 25 years

$$\text{Total capital cost} = 1913\$$$

$$\text{Energy Output} = 2000 * 0.95 = 1900 \text{ kWh/year}$$

$$\text{Electricity Cost: } \$0.05/\text{kWh} \text{ (residential electricity cost in Saudi Arabia)}$$

$$\text{Annual Energy Savings} = 1900 \text{ kWh/year} \times 0.05 \text{ USD/kWh} = 95\$$$

$$\text{Net Annual Savings} = 112 \text{ USD} - 18 \text{ USD} = 94 \$/\text{year}$$

$$\text{Payback Period} = \text{Initial cost} / \text{annual saving}$$

$$= (1913 / 94) = 20 \text{ years}$$

$$\text{Net saving} = 25 * 95 = 2350$$

$$\text{ROI} = \text{Net profit} / \text{initial cost}$$

$$((2350 - 1913) / 1913) * 100 = 18\%$$

## Conclusion

Investing in solar thermal energy systems is economically worthwhile. Beyond the financial benefits, renewable energy offers a crucial advantage: protecting the environment. By transitioning to renewables, we can sustain and develop our world while minimizing our impact on the environment and safeguarding its future.